ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

NATURAL SCIENCES

PAPER 1 · PRACTICE D

MARKS: 100 **TIME: 2 hours**

Covers all four Grade 7 strands: Life and Living (the spheres of the Earth, vertebrates and invertebrates, seed dispersal, variation), Matter and Materials (melting and boiling points, separating mixtures, red-cabbage indicator and neutralisation, metals and non-metals), Energy and Change (conduction, convection and radiation, a vacuum flask, energy chains, renewable and non-renewable energy, saving electricity) and Planet Earth and Beyond (the Sun, day and night, eclipses and early astronomers).

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Read each question carefully and answer only what is asked.
4. Write in full sentences where explanations are asked for.
5. Diagrams are NOT necessarily drawn to scale.

SECTION A

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 The layer of gases around the Earth is the …

A lithosphere. B atmosphere. C hydrosphere. D biosphere.

1.2 Which animal is an invertebrate?

A Frog B Snake C Spider D Eagle

1.3 Seeds with hooks that stick to animal fur are dispersed by …

A animals. B wind. C water. D explosion.

1.4 Which separation method is best for removing iron filings from sand?

A Filtration B Evaporation C Distillation D Using a magnet

1.5 A substance with a pH of 2 is …

A a strong acid. B neutral.
C a weak base. D a strong base.

1.6 The symbol for sodium in the periodic table is …

A So B S C Na D Sd

1.7 Heat from the Sun reaches the Earth through space by …

A conduction. B radiation. C convection. D insulation.

1.8 Which of these is a renewable energy source?

A Coal B Petrol C Natural gas D Wind

1.9 Day and night are caused by the Earth …

A spinning on its axis. B moving around the Sun.
C being tilted. D moving closer to the Moon.

1.10 During a solar eclipse, the …

A Earth is between the Sun and the Moon. B Sun is between the Earth and the Moon.
C Moon is between the Sun and the Earth. D Moon moves behind the Sun.

(10)

QUESTION 2: TRUE OR FALSE

Write TRUE or FALSE next to the question number. If the statement is FALSE, write down the correct word(s) to replace the underlined word(s).

2.1 All the water on Earth, in oceans, rivers and ice, forms the hydrosphere. (2)

2.2 Insects have eight legs. (2)

2.3 Pure water boils at 100 °C at sea level. (2)

2.4 Metals are poor conductors of electricity. (2)

2.5 The Sun is a star. (2)

(10)

TOTAL SECTION A: [20]

SECTION B

QUESTION 3: LIFE AND LIVING

3.1 Name the FOUR spheres of the Earth and give ONE example of something found in EACH. (4)

3.2 Give TWO differences between vertebrates and invertebrates. (2)

3.3 Name the class of vertebrates for EACH description: (a) has feathers and lays eggs with hard shells; (b) has dry scales and lays eggs on land; (c) feeds its young on milk. (3)

3.4 Explain why plants need to disperse (spread) their seeds away from the parent plant. (2)

3.5 Name the method of dispersal for EACH seed and give a reason: (a) a dandelion seed with a fluffy "parachute"; (b) a sweet, juicy fruit such as a marula. (4)

3.6 The heights of the learners in a Grade 7 class range from 1,35 m to 1,68 m.

3.6.1 What is variation? (2)

3.6.2 Give ONE other example of variation among learners in your class. (1)

3.6.3 Why is variation important for a species to survive? (2)

[20]

QUESTION 4: MATTER AND MATERIALS

4.1 Study the table.

Substance	Melting point (°C)	Boiling point (°C)
P	−39	357
Q	1 538	2 862
R	−218	−183

4.1.1 What is the state of EACH substance at room temperature (25 °C)? (3)

4.1.2 Which substance would be best for making a cooking pot? Give a reason. (2)

4.2 Describe how you would separate salt from salty water and explain why this method works. (3)

4.3 A learner makes an indicator by boiling red cabbage leaves in water. The indicator turns pink in an acid, purple in a neutral solution and green in a base.

4.3.1 What colour will the indicator turn in vinegar? (1)

4.3.2 What colour will it turn in a solution of soap? (1)

4.3.3 Why must the learner use a different spoon for each solution? (1)

4.4 (a) What is neutralisation? (b) Why do people take antacid tablets for heartburn? (4)

4.5 Give THREE properties of metals that make copper a good choice for electrical wires. (3)

4.6 On which side of the periodic table are most of the non-metals found? (1)

4.7 Give ONE use of a non-metal. (1)

[20]

QUESTION 5: ENERGY AND CHANGE

5.1 Name the method of heat transfer in EACH example: (a) a metal spoon in hot tea gets hot; (b) warm air rises from a heater and cold air sinks; (c) you feel the warmth of a fire from across the room. (3)

5.2 A vacuum flask keeps tea hot for many hours. Explain how EACH part reduces heat loss: (a) the vacuum between the two walls; (b) the shiny silver walls; (c) the stopper. (3)

5.3 Complete the energy chain for a coal power station: chemical potential energy in coal → _____ energy → kinetic energy of the turbine → _____ energy (2)

5.4 Give TWO advantages and TWO disadvantages of using coal to make electricity. (4)

5.5 Explain why solar energy is called a renewable energy source. (2)

5.6 Give FOUR ways to save electricity at home. (4)

5.7 Why is it dangerous to connect wires illegally to the electricity grid? (2)

[20]

QUESTION 6: PLANET EARTH AND BEYOND

6.1 Give TWO facts about the Sun. (2)

6.2 How long does it take the Earth to (a) spin once on its axis and (b) orbit the Sun once? (2)

6.3 Draw a labelled diagram to show the positions of the Sun, the Earth and the Moon during a lunar eclipse. (3)

6.4 Why must you never look directly at the Sun, even during an eclipse? (2)

6.5 Explain the difference between the geocentric model and the heliocentric model of the solar system. (4)

6.6 Name the astronomer who first suggested that the Earth moves around the Sun. (1)

6.7 Give TWO ways in which telescopes changed what people knew about space. (2)

6.8 Why does the Moon seem to shine at night? (2)

6.9 Name ONE African culture's story or use of the stars (for example to tell the seasons for planting). (2)

[20]

TOTAL SECTION B: [80]

TOTAL: 100

END OF QUESTION PAPER